Questions for Final Exam ECE 330:

Name: _____

Question #1 (6 points total):

a) Write down the Improved Two-Point Formula for first derivatives plus the order of the leading error term.
(3 points)

b) Write down the Three-Point Formula for second derivatives plus the order of the leading error term.
(3 points)

a) $f'(x) = \frac{f(x+h)-f(x-h)}{2\mathrm{h}} + O(h^2)$

b) $f''(x) = \frac{f(x+h)+f(x-h)-2\mathrm{f}(x)}{h^2} + O(h^2)$

Question #2 (13 points total):

a) Describe the procedure for solving a 2nd order differential equation step by step. Be mathematically explicit and do not just give a textual description.
(2 points)

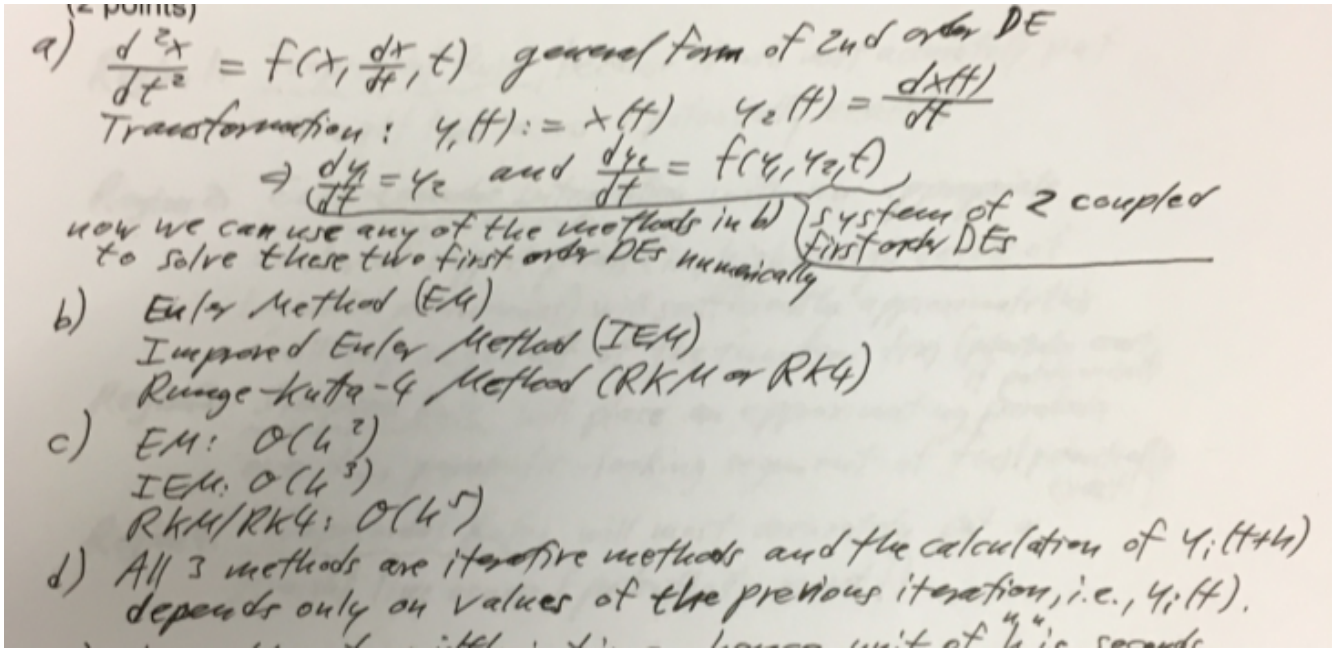
b) List three numerical methods covered in class for solving differential equations.
(3 points)

c) List the respective orders of the leading error terms for the methods above.
(3 points)

d) What does the calculation of $y_i(t+h)$ depend on in all three methods above?
(1 point)

e) What is the meaning of "h" in the three methods in b) and what is its unit?
(2 points)

f) Which of the root finding methods (Newton-Raphson, Regula Falsi, Secant, Bisection) never brackets the root?
(2 points)



f) Newton-Raphson

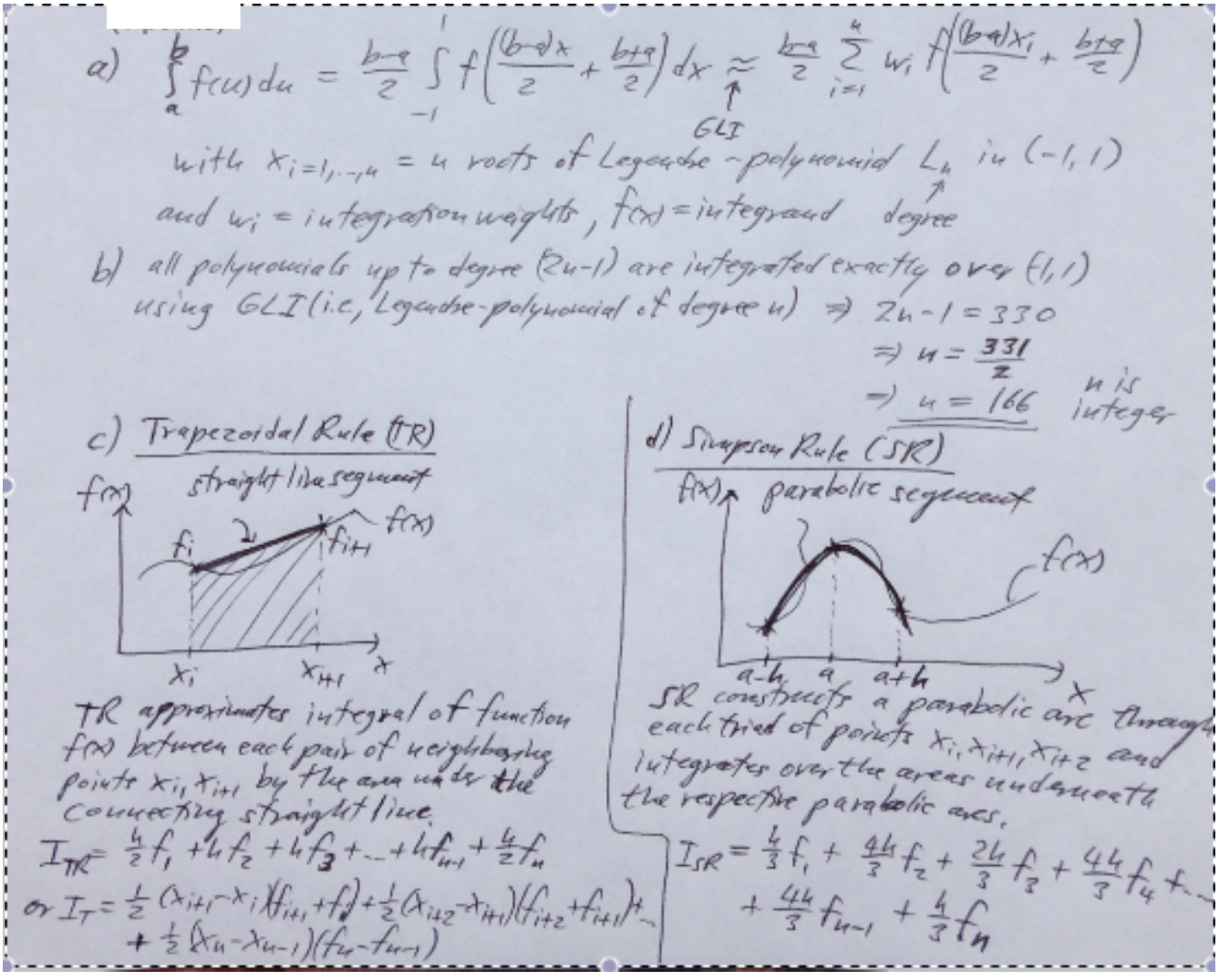
Question #3 (16 points total):

a) Write down mathematically the Gauss-Legendre Integration (GLI) approximation of an integral of a function f(x) over an arbitrary finite interval [a,b]. Explain at what locations the function f(x) is being evaluated in the GLI approximation.
(5 points)

b) If you had to integrate a polynomial $P_{330}(x)$ of degree 330, what would be the lowest degree of a Legendre polynomial to get an exact result with GLI and why?
(3 points)

c) Graphically describe the Trapezoidal Rule and explain it. List the mathematical approximation formula for the integral over several sub-intervals.
(4 points)

d) Graphically describe the Simpson Rule and explain it. List the mathematical approximation formula for the integral over several sub-intervals.
(4 points)



Question #4 (6 points total):

a) What happens to the time development of an oscillator with no driving force term if the initial conditions for the oscillator are: location x(t=0) = 0 and velocity v(t=0) = 0?
(2 points)

b) The following differential equation is given: $\frac{dy}{dt} = Ay^2 - Bt^3$ with $A, B = const.$
Derive explicitly the update equation for integrating this differential equation using the Runge-Kutta-4 Method.
(4 points)

a) because of no driving force term the system will remain at rest, i.e., will NOT start oscillating.

b)

$$\frac{dy}{dt} = f(y,t) \qquad \frac{dy}{dt} = Ay^2 - Bt^3 = f(y,t)$$

$$k^{(1)} = f(y,t) \qquad \Rightarrow k^{(1)} = Ay^2 - Bt^3$$

$$k^{(2)} = f\left(y + \frac{h}{2}k^{(1)}, t + \frac{h}{2}\right) \qquad \Rightarrow k^{(2)} = A\left[y + \frac{h}{2}k^{(1)}\right]^2 - B\left(t + \frac{h}{2}\right)^3$$

$$k^{(3)} = f\left(y + \frac{h}{2}k^{(2)}, t + \frac{h}{2}\right) \qquad \Rightarrow k^{(3)} = A\left[y + \frac{h}{2}k^{(2)}\right]^2 - B\left(t + \frac{h}{2}\right)^3$$

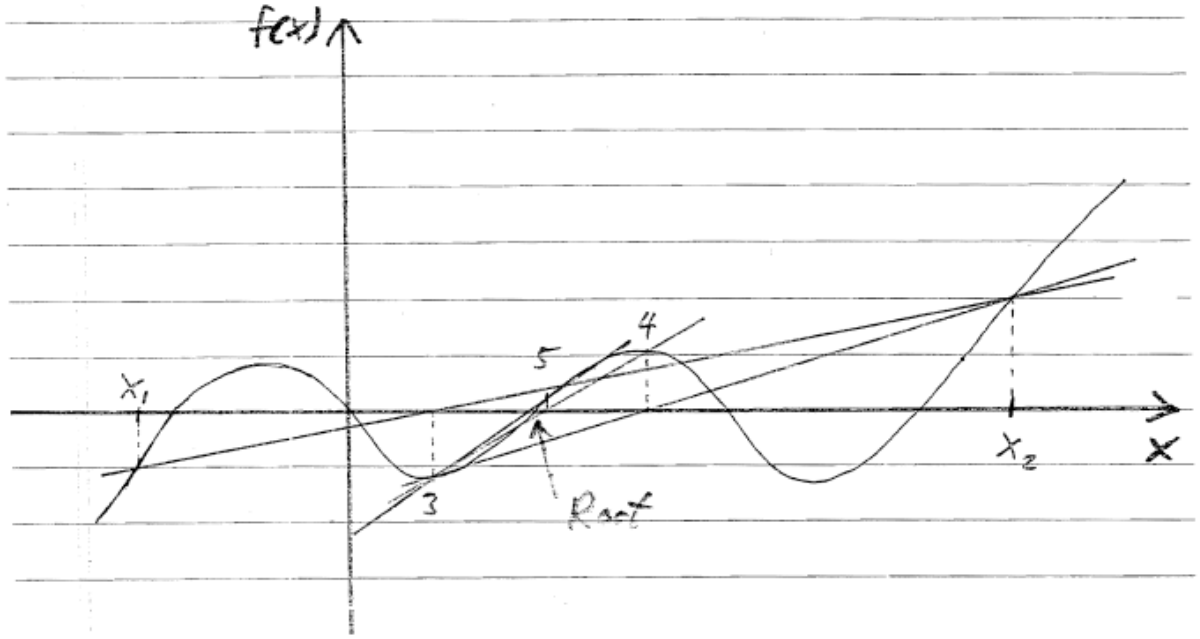
$$k^{(4)} = f\left(y + hk^{(3)}, t + h\right) \qquad \Rightarrow k^{(4)} = A\left[y + hk^{(3)}\right]^2 - B(t+h)^3$$

$$\Rightarrow y(t+h) = y(t) + h\left[\frac{k^{(1)}}{6} + \frac{k^{(2)}}{3} + \frac{k^{(3)}}{3} + \frac{k^{(4)}}{6}\right]$$

Question #5 (6 points total):

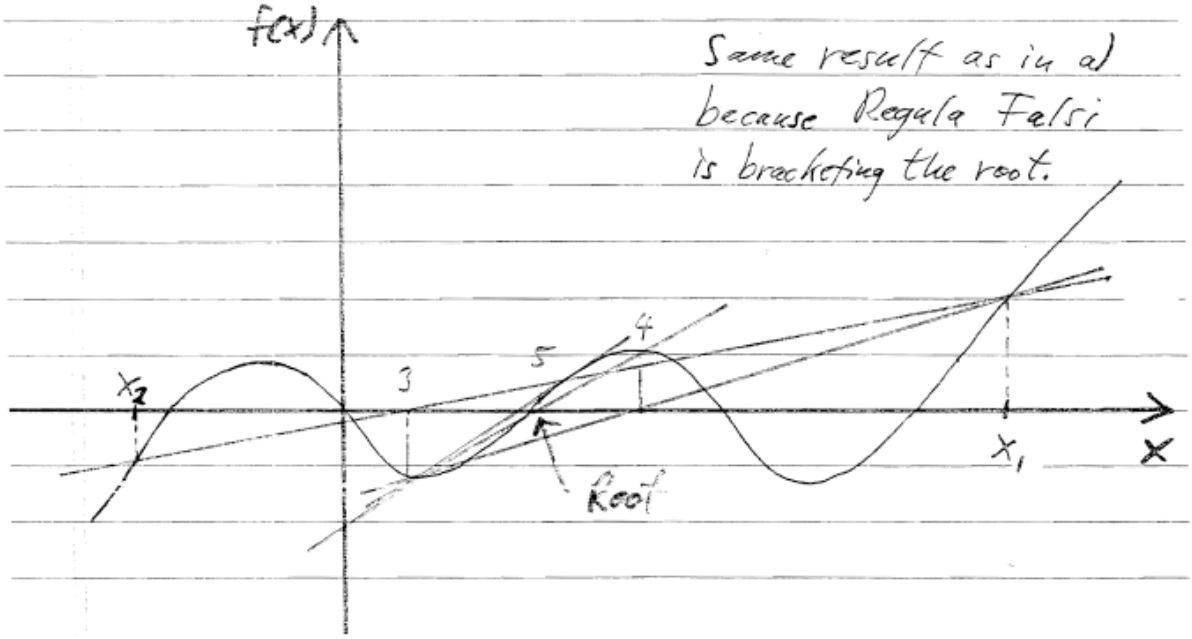
a) Applying the Regula Falsi with x_1 and x_2 as the two starting x-values, which root of the given graph will you find? Finding the correct result will require you to be extremely precise when executing the method (use of a ruler highly recommended!). Draw as many iterations as necessary to demonstrate convincingly which root will be found. NO GUESSING!

(3 points)



b) Which root will you find if x_1 and x_2 are reversed? Explain the respective results from a and b.

(3 points)



Question #6 (6 points total):

Approximate the general Laplace equation in cylindrical coordinates:

$$\nabla^2 f = \frac{\partial^2 f}{\partial r^2} + \frac{1}{r}\frac{\partial f}{\partial r} + \frac{1}{r^2}\frac{\partial^2 f}{\partial \theta^2} + \frac{\partial^2 f}{\partial z^2}$$

with the improved Two-Point and Three-Point Formulas and ignore the respective leading error terms. Mathematically reduce the result as much as possible.
(6 points)

In the following: f(z, r, theta) = PHI(z, r, phi)

$$\Delta\phi \approx \frac{\phi(z+h,r,\tau)+\phi(z-h,r,\tau)-2\phi(z,r,\tau)}{h^2}$$

$$+\frac{1}{2r}\,\frac{\phi(z,r+h,\tau)-\phi(z,r-h,\tau)}{h}$$

$$+\frac{\phi(z,r+h,\tau)+\phi(z,r-h,\tau)-2\phi(z,r,\tau)}{h^2}$$

$$+\frac{1}{r^2}\,\frac{\phi(z,r,\tau+h)+\phi(z,r,\tau-h)-2\phi(z,r,\tau)}{h^2} = 0 \quad \Big| \cdot h^2$$

$$\Rightarrow \phi(z+h,r,\tau)+\phi(z-h,r,\tau)+\phi(z,r+h,\tau)+\phi(z,r-h,\tau)$$

$$-4\phi(z,r,\tau)+\frac{h}{2r}\left[\phi(z,r+h,\tau)-\phi(z,r-h,\tau)\right]$$

$$+\frac{1}{r^2}\left[\phi(z,r,\tau+h)+\phi(z,r,\tau-h)-2\phi(z,r,\tau)\right]$$

$$= \phi(z+h,r,\tau)+\phi(z-h,r,\tau)+\left(1+\frac{h}{2r}\right)\phi(z,r+h,\tau)$$

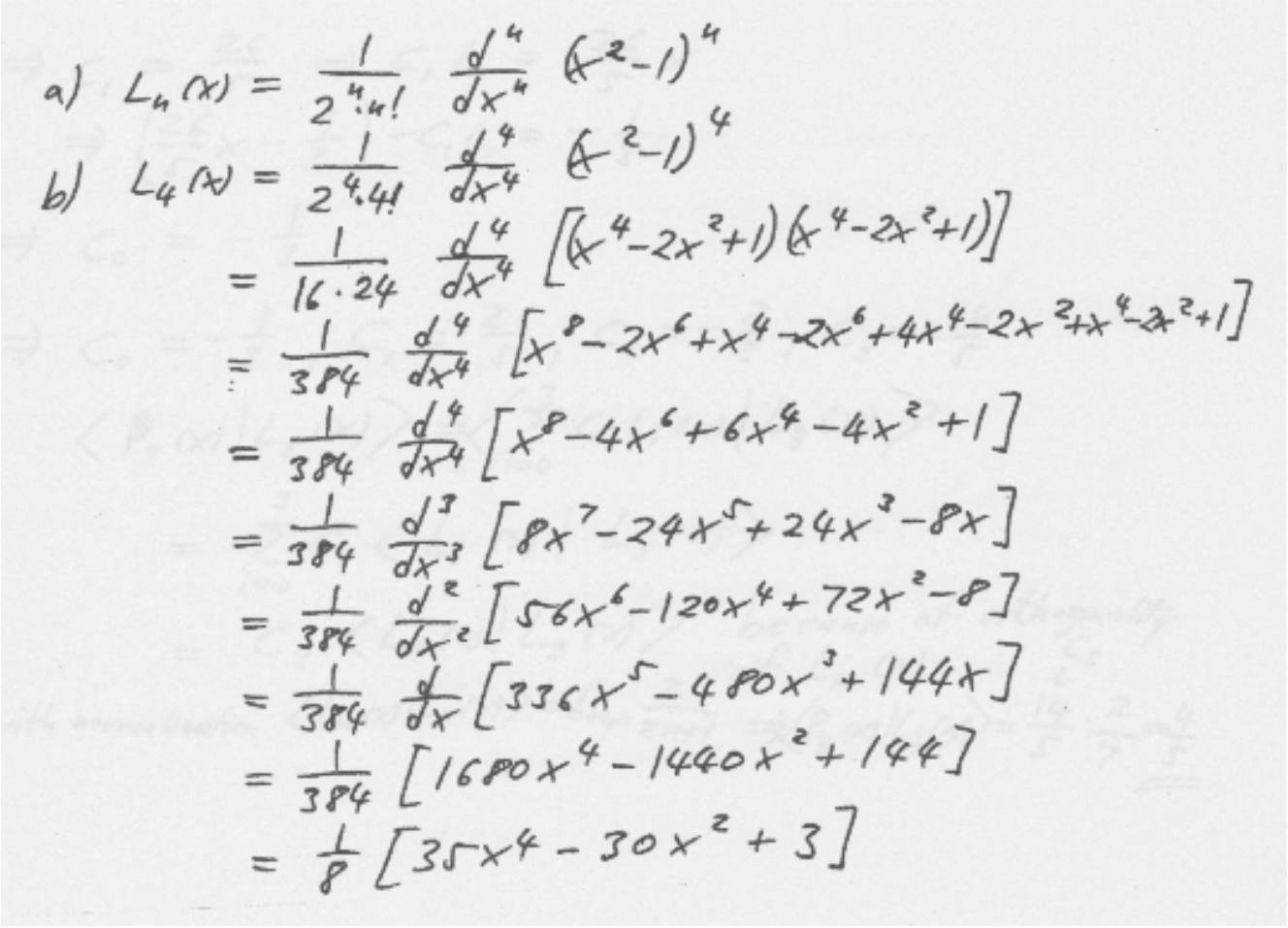
$$+\left(1-\frac{h}{2r}\right)\phi(z,r-h,\tau)-\left(4+\frac{2}{r^2}\right)\phi(z,r,\tau)$$

$$+\frac{1}{r^2}\left[\phi(z,r,\tau+h)+\phi(z,r,\tau-h)\right] = 0$$

Question #7 (8 points total):

a) List Rodrigues formula for Legendre Polynomials.
(2 points)

b) For the polynomial $P_4(x) = -x^4 - \frac{35}{4}x^3 - 17\mathrm{x}^2 - 330\mathrm{x} + \frac{49}{37}$: calculate the explicit value of $\langle P_4(x) \mid L_4(x) \rangle$.
(6 points)



So, if $P_4(x)$ were to be expanded into Legendre polynomials, the coefficient for the x^4-term would be $C_4 = -\frac{8}{35}$.

Therefore, $\langle P_4(x) \mid L_4(x) \rangle = \langle C_4 L_4(x) \mid L_4(x) \rangle$ because of *orthogonality*

and $\langle L_4(x) \mid L_4(x) \rangle = \frac{2}{2n+1} = \frac{2}{9}$ because of *normalization*

so $\langle P_4(x) \mid L_4(x) \rangle = C_4 \frac{2}{9} = -\frac{16}{315}$

Question #8 (6 points total):

Describe the Simulated Annealing (SA) algorithm in detailed pseudo-code form. (6 points)

```
Set Temperature T = T0
Initial parameter vector P
Calculate initial current fitness/energy E_current
Set best fitness/energy E_best = E_current
Set best parameter vector P_best = P
Set cooling rate: 0 < lambda < 1

LOOP:
{
    Modify parameter vector P to yield P'
    Calculate resulting temporary fitness/energy: E_temp

    if (E_temp< E_current)
    {
        E_current = E_temp
        make P' the new P: P = P'
        if (E_current < E_best)
        {
            E_best = E_current
            P_best = P
            if (E_best < user-defined fitness/energy)
            {
                STOP optimization
            }
        }
    }
    else
    {
        if (random number [0...1] < exp[-(E_temp - E_best)/T])      [COMMENT: accept unfavorable step with Boltzmann-probability [0...1]]
        {
            E_current = E_temp
            make P' the new P: P = P'
        }
        else
        {
            reject P' and go back to P
        }
    }
    T = lambda * T      [COMMENT: cool temperature for Boltzmann-probability, i.e., make it harder to accept unfavorable steps]
    if (T < user-defined minimum T)
    {
        STOP optimization
    }
    else
    {
        goto LOOP
    }
}
```

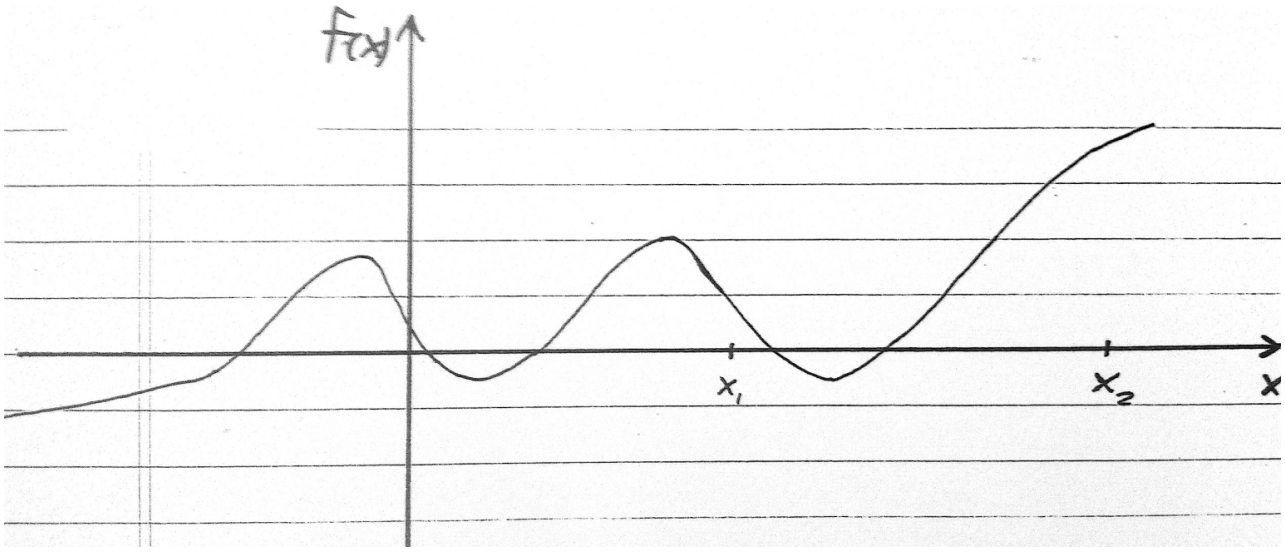
} exit criterion (I)

} exit criterion (II)

can also be max. number of allowed/ prescribed iterations exhausted/exceeded

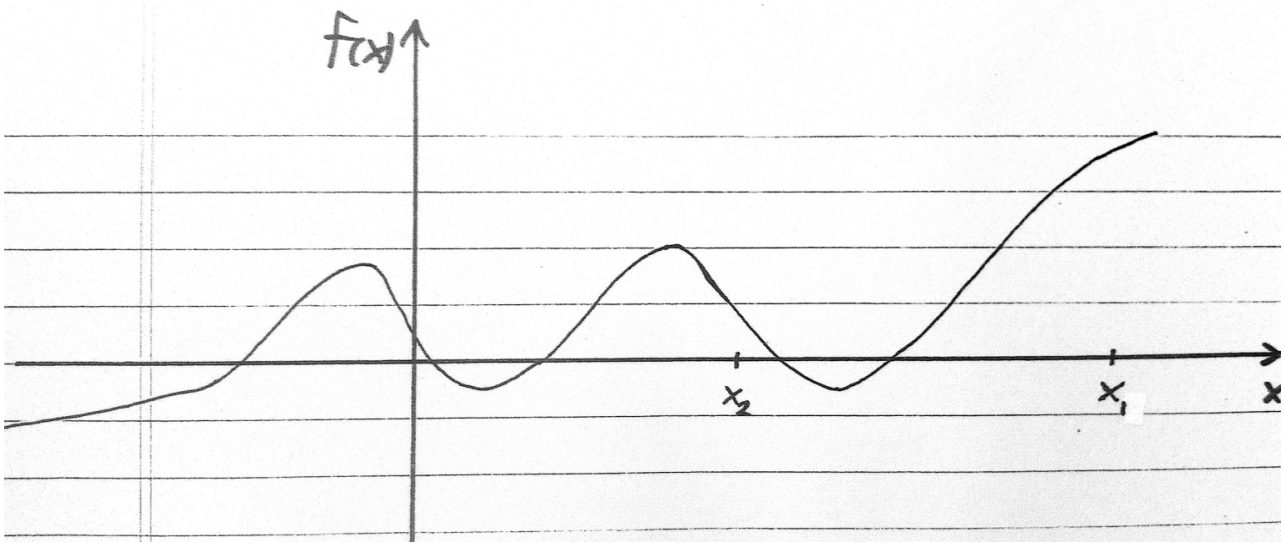
Question #9 (4 points total):

a) Applying the Bisection Method with x_1 and x_2 as the two starting x-values, which root of the given graph will you find? Finding the correct result will require you to be extremely precise when executing the method (use of a ruler highly recommended!). Draw as many iterations as necessary to demonstrate convincingly which root will be found. NO GUESSING!
(2 points)



Bisection Method not applicable since it has to bracket the root, i.e., $f(x_1)*f(x_2)<0$. Not the case here!

b) Which root will you find if x_1 and x_2 are reversed? Explain the respective results from a and b.
(2 points)



Bisection Method not applicable since it has to bracket the root, i.e., $f(x_1)*f(x_2)<0$. Not the case here!

Question #10 (9 points total):

a) You have the following parameter vector in the population of a Genetic Algorithm:
$[1 \quad -5 \quad -1 \quad 4 \quad 9 \quad 3 \quad -7 \quad 8]$
Apply the *inversion* operator and write down the resulting parameter vector.
(1 points)

b) The following two children have resulted from a *crossover* operation in a Genetic Algorithm:

Child 1: $[2 \quad -3 \quad 1 \quad -1 \quad 5 \quad 3 \quad -7 \quad -8]$

Child 2: $[8 \quad 2 \quad 7 \quad 6 \quad 4 \quad 5 \quad -9 \quad 0]$

The crossover area started at position 5 (including) and ended at position 8 (including). What were the initial parent parameter vectors?
(4 points)

c) The following two parents have been selected for a *crossover* operation in a Genetic Algorithm for a Traveling Salesman Problem:

Parent 1: $[1 \quad 3 \quad 5 \quad 2 \quad 7 \quad 6 \quad 4 \quad 8]$

Parent 2: $[8 \quad 2 \quad 7 \quad 1 \quad 6 \quad 3 \quad 4 \quad 5]$

The crossover area starts at position 3 (including) and ends at position 6 (including). Which two resulting offspring parameter vectors will result? Will there be a problem with the offspring parameter vectors in this case or not? Justify your answer in any case.
(4 points)

a) $[8 \quad -7 \quad 3 \quad 9 \quad 4 \quad -1 \quad -5 \quad 1]$

b) Child 1: $[2 \quad -3 \quad 1 \quad -1 \quad 5 \quad 3 \quad -7 \quad -8]$
[----------]

Child 2: $[8 \quad 2 \quad 7 \quad 6 \quad 4 \quad 5 \quad -9 \quad 0]$
[----------]

Parent 1: $[2 \quad -3 \quad 1 \quad -1 \quad 4 \quad 5 \quad -9 \quad 0]$

Parent 2: $[8 \quad 2 \quad 7 \quad 6 \quad 5 \quad 3 \quad -7 \quad -8]$

c) Parent 1: $[1 \quad 3 \quad 5 \quad 2 \quad 7 \quad 6 \quad 4 \quad 8]$
[----------]

Parent 2: $[8 \quad 2 \quad 7 \quad 1 \quad 6 \quad 3 \quad 4 \quad 5]$
[----------]

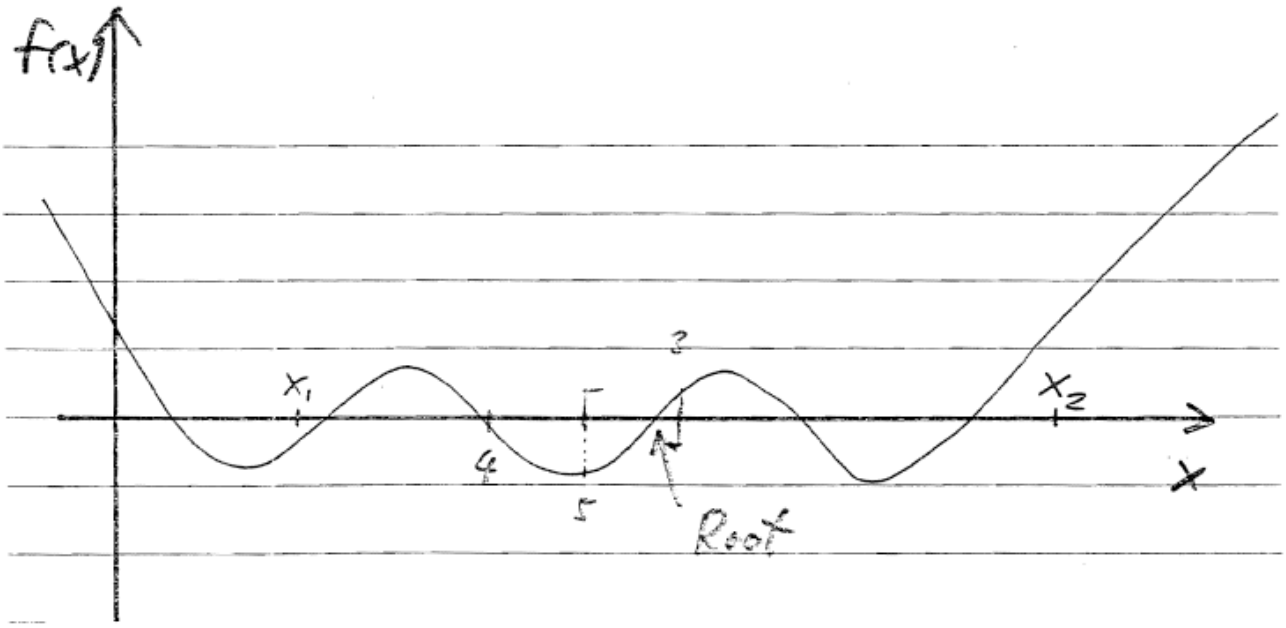
Child 1: $[1 \quad 3 \quad 7 \quad 1 \quad 6 \quad 3 \quad 4 \quad 8]$

Child 2: $[8 \quad 2 \quad 5 \quad 2 \quad 7 \quad 6 \quad 4 \quad 5]$

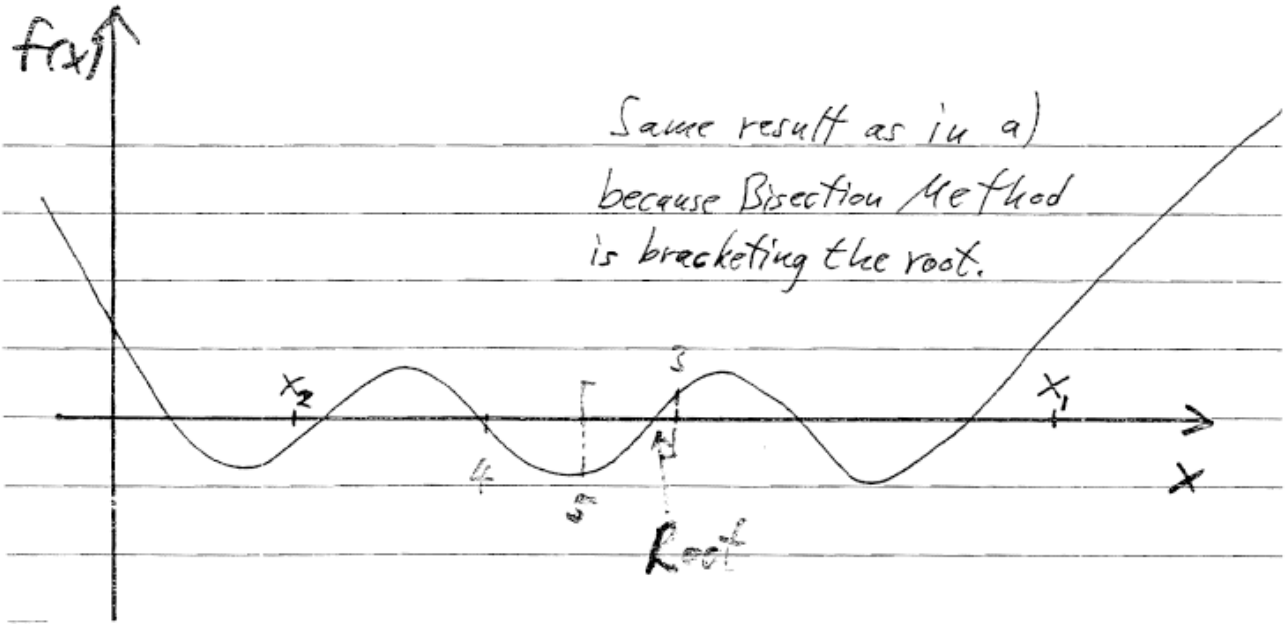
Both offspring parameter vectors (children) are *invalid* for the TSP problem because of both duplicate and missing city numbers.

Question #11 (6 points total):

a) Applying the Bisection Method with x_1 and x_2 as the two starting x-values, which root of the given graph will you find? Finding the correct result will require you to be extremely precise when executing the method (use of a ruler highly recommended!). Draw as many iterations as necessary to demonstrate convincingly which root will be found. NO GUESSING!
(3 points)



b) Which root will you find if x_1 and x_2 are reversed? Explain the result from a and b.
(3 points)



Question #12 (10 points total):

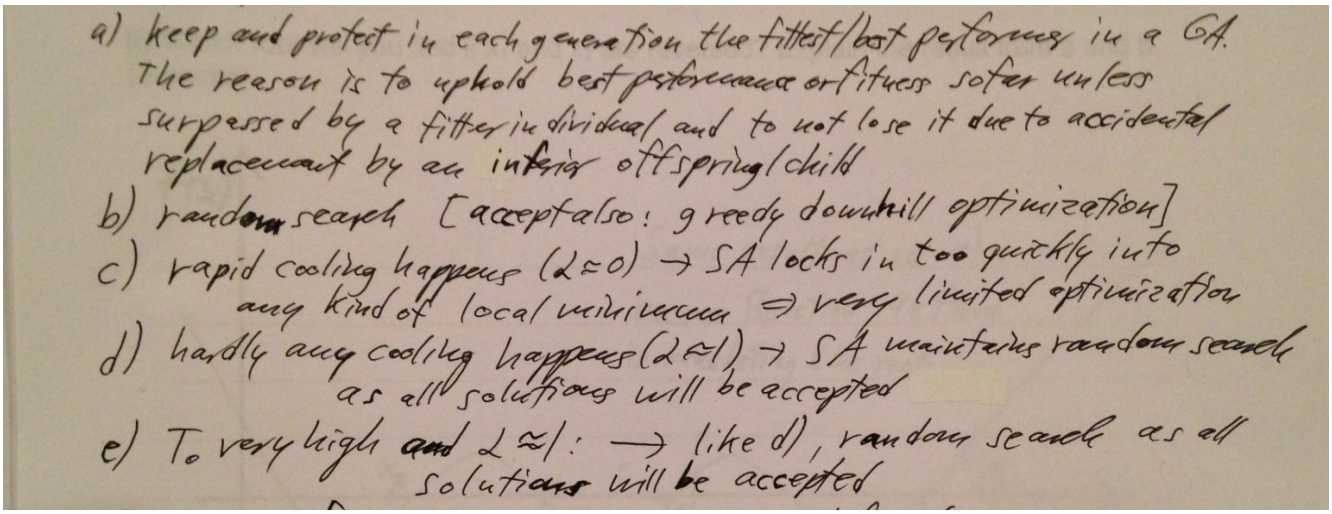
a) Explain the concept of the Elitist approach in Genetic Algorithms. Why is it important?
(2 points)

b) What does Simulated Annealing turn into when you eliminate the section of the algorithm that accepts a worse performing solution via the Boltzmann-probability?
(2 point)

c) What happens to Simulated Annealing when the cooling rate Lambda is very close to 0?
(2 points)

d) What happens to Simulated Annealing when the cooling rate Lambda is very close to 1?
(2 points)

e) What happens to Simulated Annealing when the initial temperature is very high and the cooling rate Lambda very close to 1?
(2 points)



Question #13 (8 points total):

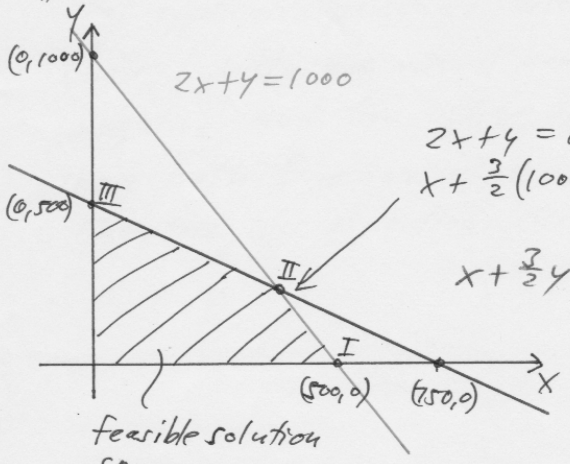
A store has requested a manufacturer to produce pants and sports jackets. The manufacturer has 750 m^2 of cotton textile and 1,000 m^2 of polyester. Every pant requires 1 m^2 of cotton and 2 m^2 of polyester. Every jacket requires 1.5 m^2 of cotton and 1 m^2 of polyester. The price of the pants is \$50 and of the jacket \$40. What is the number of pants and jackets that the manufacturer has to deliver to the store so that these items yield the maximum sale?
(8 points)

1) define unknowns:
x = # of pants
y = # of jackets

2) Write objective function:
$f(x,y) = 50x + 40y$

3) Write constraints as system of inequalities:
$x + \frac{3}{2}y \leq 750$
$2x + y \leq 1000$
$x \geq 0$
$y \geq 0$
$x, y \in \mathbb{N}_0$

4) find set of feasible solutions:



$2x + y = 1000 \Rightarrow y = 1000 - 2x$
$x + \frac{3}{2}(1000 - 2x) = -2x + 1500 = 750 \Rightarrow \boxed{x = 375}$
$\Rightarrow \boxed{y = 250}$

$x + \frac{3}{2}y = 750$

5) Calculate coordinates of valid vertices:
I: (500, 0)
II: (375, 250)
III: (0, 500)

6) Calculate objective function value at each vertex:
I: $f(500,0) = 25,000$
II: $f(375,250) = 18,750 + 10,000 = 28,750$
III: $f(0,500) = 20,000$

$\Rightarrow$ Answer: 375 pants and 250 jackets